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Subject : AIML

Assignment 2

Prolog (Programming in Logic) is a programming language used for Artificial Intelligence (AI) and rule-based systems.

The Smart Street Light Agent is a simple rule-based AI system developed in Prolog. The system automatically controls street lights based on the current time and motion detection. It first checks whether it is day, evening, or night and then checks if any motion is present in that area. When motion is detected at night, the light is switched to full brightness. If no motion is detected at night, the light is kept in dim mode to save electricity. During the evening, the light is set to medium brightness when motion is present, and during the daytime the light is turned off automatically. The program uses facts to store sensor information and rules to make decisions, demonstrating how Prolog performs intelligent reasoning for a smart lighting application

The screenshot shows the SWISH Prolog IDE interface. The left pane contains the Prolog code for the Smart Street Light Agent, and the right pane shows the execution results for five agents.

```
1 % Smart Street Light Agent
2 % Facts
3 % light(Area, Time, Motion)
4 light(areal, night, motion).
5 light(areal, night, no_motion).
6 light(areal, evening, motion).
7 light(areal, day, motion).
8 light(areal, day, no_motion).
9
10 % Rules
11 % Night and motion detected -> Full brightness
12 action(Area, full_brightness) :-
13     light(Area, night, motion).
14
15 % Night and no motion -> Dim Light
16 action(Area, dim_light) :-
17     light(Area, night, no_motion).
18
19 % Evening and motion -> Medium brightness
20 action(Area, medium_brightness) :-
21     light(Area, evening, motion).
22
23 % Day time -> Light OFF
24 action(Area, turn_light_off) :-
25     light(Area, day, _).
26 % Display result
27 agent(Area) :-
28     action(Area, Action),
29     write('Area: '), write(Area),
30     write(' -> Action: '), write(Action),
31     nl.
```

The right pane displays the execution results for five agents:

- agent(areal):** Area: areal -> Action: full_brightness, true
- agent(areal):** Area: areal -> Action: dim_light, true
- agent(areal):** Area: areal -> Action: medium_brightness, true
- agent(areal):** Area: areal -> Action: turn_light_off, true
- agent(areal):** Area: areal -> Action: turn_light_off, true

The bottom of the right pane shows a search bar and a "Run" button.